# Chapter 2: Atomic Structure & Bonding

## Key Equations

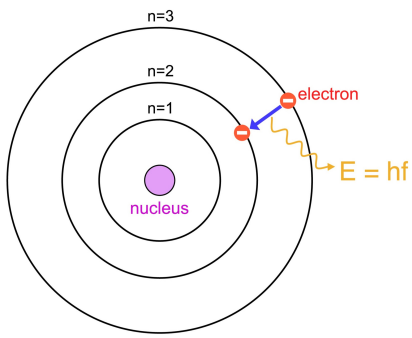
**Average Atomic Weight:** $A_{avg} = \sum f_i A_i$
($A_{avg}$: average atomic wt, $f_i$: fraction of occurrence, $A_i$: atomic wt of isotope $i$)

**Force of Attraction:** $F_A = \frac{|Z_1 Z_2| e^2}{4\pi\epsilon_0 r^2}$
($F_A$: attractive force, $Z_1, Z_2$: ion valences, $r$: interatomic distance)
Constants: $e = 1.602 \times 10^{-19}$ C (electron charge), $\epsilon_0 = 8.85 \times 10^{-12}$ F/m (permittivity of vacuum).

**Percent Ionic Character (%IC):**
$\%IC = \left(1 - e^{-0.25(X_A - X_B)^2}\right) \times 100$
($X_A, X_B$: electronegativities of elements A and B)

## Concepts

**Bohr Model:** Electrons orbit a central nucleus in discrete energy levels.




**Bonding:** Ionic (electron transfer), Covalent (electron sharing), Metallic (sea of delocalized electrons).

**Example: Force of Attraction**
Calculate force between $K^+$ ($Z_1 = +1$) and $I^-$ ($Z_2 = -1$) separated by 0.314 nm.
$r = 0.314 \times 10^{-9}$ m.
$F_A = \frac{|(1)(-1)|(1.602\times10^{-19})^2}{4\pi(8.85\times10^{-12})(0.314\times10^{-9})^2} = 2.33 \times 10^{-9}$ N

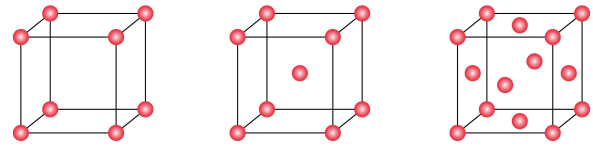
# Chapter 3: Crystal Structures

## Key Equations & Geometry

**SC Edge Length:** $a = 2R$, $n = 1$ atom/cell.
**BCC Edge Length:** $a = \frac{4R}{\sqrt{3}}$, $n = 2$ atoms/cell.
**FCC Edge Length:** $a = 2R\sqrt{2}$, $n = 4$ atoms/cell.
($a$: unit cell edge length, $R$: atomic radius, $n$: atoms per unit cell)

**Atomic Packing Factor (APF):** Fraction of solid sphere vol.
$APF = \frac{\text{Vol. of atoms}}{\text{Vol. of unit cell}} = \frac{n(\frac{4}{3}\pi R^3)}{a^3}$
(SC = 0.52, BCC = 0.68, FCC = 0.74)

**Unit Cell Volume:** $V_c = a^3$
**Theoretical Density:** $\rho = \frac{nA}{V_c N_A}$
Constant: $N_A = 6.022 \times 10^{23}$ atoms/mol (Avogadro's number).




## Coordinate Systems & Miller Indices

**Points:** $q\ r\ s$ (fractional multiples of edge lengths).
**Directions:** $[u\ v\ w]$ (Head coord. - Tail coord., clear fractions).
**Planes (Miller Indices):** 1. If plane passes through origin, establish new origin. 2. Find intercepts on $x, y, z$ axes (parallel $= \infty$). 3. Take reciprocals of intercepts. 4. Multiply to clear fractions (smallest integers). 5. Enclose in parentheses $(hkl)$. Negative intercepts get a bar (e.g., $\bar{1}$).

## Concepts

**Crystalline vs. Amorphous:** Crystalline has long-range order; amorphous lacks it.
**Isotropy/Anisotropy:** Isotropic properties are independent of direction; anisotropic vary.

# Chapter 4: Imperfections in Solids

## Concepts

**Vacancy Defect:** A point defect where an atom is missing from its normal structural lattice site.
**Self-Interstitial Defect:** A point defect where an extra atom from the crystal is crowded into an interstitial void space that is normally empty.

## Key Equations

**Total Atomic Sites:** $N = \frac{N_A \rho}{A}$
($N$: total atomic sites per volume, $\rho$: density)

**Equilibrium Vacancies:** $N_v = N \exp\left(-\frac{Q_v}{kT}\right)$
($N_v$: number of vacancies, $Q_v$: activation energy for vacancy formation)
Constant: $k = 8.62 \times 10^{-5}$ eV/atom $\cdot$ K. ($T$: temp in Kelvin).

**Mass to Weight %:** $C_1 = \frac{m_1}{m_1 + m_2} \times 100$

**Weight % to Atom %:**
$C_1' = \frac{C_1/A_1}{C_1/A_1 + C_2/A_2} \times 100$
($C_1'$: atom percent of element 1, $A_1, A_2$: atomic weights)

**Atom % to Weight %:**
$C_1 = \frac{C_1' A_1}{C_1' A_1 + C_2' A_2} \times 100$
($C_1$: weight percent of element 1)

**Example: Equilibrium Vacancies**
Gold at 900°C (1173 K), $Q_v = 0.98$ eV/atom, $A = 196.9$ g/mol, $\rho = 18.63 \times 10^6$ g/m$^3$.
$N = \frac{(6.022\times10^{23})(18.63\times10^6)}{196.9} = 5.70 \times 10^{28}$ atoms/m$^3$.
$N_v = (5.70 \times 10^{28}) \exp\left(-\frac{0.98}{(8.62\times10^{-5})(1173)}\right)$
$= 3.52 \times 10^{24}$ vacancies/m$^3$